Tarea M9 – Yamil Salomón

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Práctica: Operaciones con más de una tabla

1) Nombres de empleados y títulos de puesto (INNER JOIN)

```sql
USE AdventureWorks2022;

-- Ejercicio 1: Obtener nombres y puestos con INNER JOIN
SELECT
    p.FirstName,
    p.LastName,
    e.JobTitle
FROM HumanResources.Employee e
INNER JOIN Person.Person p
    ON e.BusinessEntityID = p.BusinessEntityID;
```

	FirstName	LastName	JobTitle
1	Michael	Sullivan	Senior Design Engineer
2	Peter	Krebs	Production Control Manager
3	Jo	Brown	Production Supervisor - WC60
4	Guy	Gilbert	Production Technician - WC60
5	Janice	Galvin	Tool Designer
6	Thierry	D'Hers	Tool Designer
7	Ovidiu	Cracium	Senior Tool Designer
8	Michael	Raheem	Research and Development Manager
9	Gigi	Matthew	Research and Development Engineer
10	Diane	Margheim	Research and Development Engineer
11	Dylan	Miller	Research and Development Manager
12	Jossef	Goldberg	Design Engineer
13	Gail	Erickson	Design Engineer
14	Rob	Walters	Senior Tool Designer
15	Roberto	Tamburello	Engineering Manager
16	JoLynn	Dobney	Production Supervisor - WC60
17	Bryan	Baker	Production Technician - WC60
18	James	Kramer	Production Technician - WC60
19	Terri	Duffy	Vice President of Engineering
20	Ken	Sánchez	Chief Executive Officer
21	James	Hamilton	Vice President of Production
22	Jill	Williams	Marketing Specialist
23	Mary	Gibson	Marketing Specialist

2) Nombres de clientes (LEFT JOIN)

```sql
    -- Ejercicio 2: Obtener nombres de todos los clientes con LEFT JOIN
SELECT
    c.CustomerID,
    p.FirstName,
    p.LastName
FROM Sales.Customer c
LEFT JOIN Person.Person p
    ON c.PersonID = p.BusinessEntityID;
```

	CustomerID	FirstName	LastName
1	29485	Catherine	Abel
2	29486	Kim	Abercrombie
3	29487	Humberto	Acevedo
4	29484	Gustavo	Achong
5	29488	Pilar	Ackerman
6	28866	Aaron	Adams
7	13323	Adam	Adams
8	21139	Alex	Adams
9	29170	Alexandra	Adams
10	19419	Allison	Adams
11	11971	Amanda	Adams
12	26746	Amber	Adams
13	16845	Andrea	Adams
14	18504	Angel	Adams
15	13280	Bailey	Adams
16	28678	Ben	Adams
17	18646	Blake	Adams
18	29491	Carla	Adams
19	19822	Carlos	Adams
20	26598	Charles	Adams
21	19410	Chloe	Adams
22	21423	Connor	Adams
23	18075	Courtney	Adams

3) Órdenes y Clientes (RIGHT JOIN)

```sql
    -- Ejercicio 3: Órdenes y Clientes asegurando traer todos los clientes con RIGHT JOIN
SELECT
    soh.SalesOrderID,
    c.CustomerID
FROM Sales.SalesOrderHeader soh
RIGHT JOIN Sales.Customer c
    ON soh.CustomerID = c.CustomerID;
```

	SalesOrderID	CustomerID
1	NULL	1
2	NULL	2
3	NULL	3
4	NULL	4
5	NULL	5
6	NULL	6
7	NULL	7
8	NULL	8
9	NULL	9
10	NULL	10
11	NULL	11
12	NULL	12
13	NULL	13
14	NULL	14
15	NULL	15
16	NULL	16
17	NULL	17
18	NULL	18
19	NULL	19
20	NULL	20
21	NULL	21
22	NULL	22
23	NULL	23

4) Productos y Reseñas (FULL OUTER JOIN)

```sql
    -- Ejercicio 4: Productos y Reseñas asegurando traer ambos lados con FULL OUTER JOIN
SELECT
    p.Name,
    pr.Comments
FROM Production.Product p
FULL OUTER JOIN Production.ProductReview pr
    ON p.ProductID = pr.ProductID;
```

	Name	Comments
1	Adjustable Race	NULL
2	Bearing Ball	NULL
3	BB Ball Bearing	NULL
4	Headset Ball Bearings	NULL
5	Blade	NULL
6	LL Crankarm	NULL
7	ML Crankarm	NULL
8	HL Crankarm	NULL
9	Chainring Bolts	NULL
10	Chainring Nut	NULL
11	Chainring	NULL
12	Crown Race	NULL
13	Chain Stays	NULL
14	Decal 1	NULL
15	Decal 2	NULL
16	Down Tube	NULL
17	Mountain End Caps	NULL
18	Road End Caps	NULL
19	Touring End Caps	NULL
20	Fork End	NULL
21	Freewheel	NULL
22	Flat Washer 1	NULL
23	Flat Washer 6	NULL

5) Combinación de productos y categorías (CROSS JOIN)

```sql
-- Ejercicio 5: Combinación total de productos y categorías usando CROSS JOIN
SELECT
    p.Name AS ProductName,
    pc.Name AS CategoryName
FROM Production.Product p
CROSS JOIN Production.ProductCategory pc;
```

	ProductName	CategoryName
1	Adjustable Race	Accessories
2	All-Purpose Bike Stand	Accessories
3	AWC Logo Cap	Accessories
4	BB Ball Bearing	Accessories
5	Bearing Ball	Accessories
6	Bike Wash - Dissolver	Accessories
7	Blade	Accessories
8	Cable Lock	Accessories
9	Chain	Accessories
10	Chain Stays	Accessories
11	Chainring	Accessories
12	Chainring Bolts	Accessories
13	Chainring Nut	Accessories
14	Classic Vest, L	Accessories
15	Classic Vest, M	Accessories
16	Classic Vest, S	Accessories
17	Cone-Shaped Race	Accessories
18	Crown Race	Accessories
19	Cup-Shaped Race	Accessories
20	Decal 1	Accessories
21	Decal 2	Accessories
22	Down Tube	Accessories
23	External Lock Washer 1	Accessories

6) Combinar nombres de productos y modelos (UNION vs UNION ALL)

```sql
-- Ejercicio 6a: Combinar nombres eliminando duplicados (UNION)
SELECT Name
FROM Production.Product
UNION
SELECT Name
FROM Production.ProductModel;

-- Ejercicio 6b: Combinar nombres manteniendo todos los registros (UNION ALL)
SELECT Name
FROM Production.Product
UNION ALL
SELECT Name
FROM Production.ProductModel;
```

Results | Messages

	Name
1	Adjustable Race
2	All-Purpose Bike Stand
3	AWC Logo Cap
4	BB Ball Bearing
5	Bearing Ball
6	Bike Wash
7	Bike Wash - Dissolver
8	Blade
9	Cable Lock
10	Chain
11	Chain Stays

	Name
1	Adjustable Race
2	All-Purpose Bike Stand
3	AWC Logo Cap
4	BB Ball Bearing
5	Bearing Ball
6	Bike Wash - Dissolver
7	Blade
8	Cable Lock
9	Chain
10	Chain Stays
11	Chainring

7) Clasificación de puestos con CASE y COALESCE

```sql
-- Ejercicio 7: Clasificación condicional de puestos con CASE y protección contra nulos con COALESCE
SELECT
    BusinessEntityID,
    COALESCE(JobTitle, 'No Title') AS JobTitle,
    CASE
        WHEN JobTitle LIKE '%Manager%' THEN 'Manager'
        ELSE 'Not Manager'
    END AS JobCategory
FROM HumanResources.Employee;
```

	BusinessEntityID	JobTitle	JobCategory
1	1	Chief Executive Officer	Not Manager
2	2	Vice President of Engineering	Not Manager
3	3	Engineering Manager	Manager
4	4	Senior Tool Designer	Not Manager
5	5	Design Engineer	Not Manager
6	6	Design Engineer	Not Manager
7	7	Research and Development Manager	Manager
8	8	Research and Development Engineer	Not Manager
9	9	Research and Development Engineer	Not Manager
10	10	Research and Development Manager	Manager
11	11	Senior Tool Designer	Not Manager
12	12	Tool Designer	Not Manager
13	13	Tool Designer	Not Manager
14	14	Senior Design Engineer	Not Manager
15	15	Design Engineer	Not Manager
16	16	Marketing Manager	Manager
17	17	Marketing Assistant	Not Manager
18	18	Marketing Specialist	Not Manager
19	19	Marketing Assistant	Not Manager
20	20	Marketing Assistant	Not Manager
21	21	Marketing Specialist	Not Manager
22	22	Marketing Specialist	Not Manager
23	23	Marketing Specialist	Not Manager

8) Evaluación de Cuotas de Venta (ISNULL)

```sql
-- Ejercicio 8: Identificar cuotas de ventas nulas usando ISNULL y CAST
SELECT
    BusinessEntityID,
    SalesQuota,
    ISNULL(CAST(SalesQuota AS VARCHAR), 'No Quota') AS QuotaStatus
FROM Sales.SalesPerson;
```

	BusinessEntityID	SalesQuota	QuotaStatus
1	274	NULL	No Quota
2	275	300000.00	300000.00
3	276	250000.00	250000.00
4	277	250000.00	250000.00
5	278	250000.00	250000.00
6	279	300000.00	300000.00
7	280	250000.00	250000.00
8	281	250000.00	250000.00
9	282	250000.00	250000.00
10	283	250000.00	250000.00
11	284	300000.00	300000.00
12	285	NULL	No Quota
13	286	250000.00	250000.00
14	287	NULL	No Quota
15	288	250000.00	250000.00
16	289	250000.00	250000.00
17	290	250000.00	250000.00